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Review

Reviewed Work(s): What Algorithms Want: Imagination in the Age of Computing by Ed Finn

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automata capable of serving tea and performing on dramatic scenes. Less a fault of the book, these omissions are better seen as opportunities to supplement Robots with other historically-focused readings on a course syllabus. Yet in sum, both students of the humanities, looking to better understand the technology of robotics, and technical students, looking to understand the social issues surrounding robots, will be well-served by Robots.

Colin Garvey, Troy NY, USA

Ed Finn, *What Algorithms Want: Imagination in the Age of Computing*, Cambridge MA: The MIT Press, 2017, 258 pages, ISBN 9780262035927, $29.95.

Ed Finn's *What Algorithms Want* shows the power that algorithms have in today's world, to the point where huge companies like Google or Facebook can be considered as simply being the necessary embodiment of structures surrounding some core algorithms.

In the beginning of the book, Finn analyzes what an algorithm is, with the analysis going much deeper than conventional definitions of algorithms as a sequence of computing steps. Starting from the obvious fact that an algorithm is expressed in a language, he draws analogies ranging from Genesis' logos to mystical incantations to fictional cyberpunk devices, and states that algorithms, as the tool of computation, can be considered magical. Besides the pragmatic definition of an algorithm as a method for solving a problem, the author invites us to consider algorithms from other aspects, extending across disciplines (cybernetics, symbolic language, and technical cognition), and to recognize our daily interactions with them, in order to arrive to a cultural understanding of algorithms.

Finn then examines in detail a handful of algorithms being used on a massive scale in today's world. The first one is the case of intelligent assistants, like Apple's Siri, and of Google's search engine, where the common functional requirement is to provide a service that understands the needs of the user and reacts accordingly. Finn draws analogies to Diderot's Enlightenment project of the Encyclopédie and to Wikipedia, two other

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projects of collecting all human knowledge, as well the computer in Star Trek or the movie *Her*.

The second case examined is Netflix's recommendation system, where, based on one's viewing history and an extended taxonomy of content, an algorithm suggests other content that would satisfy the viewer. The next step for such a system was to use it for the creation of new content that would appeal to viewers, and Finn explains the steps taken towards producing House of Cards and how, according to him, this algorithmic production is visible in the end product.

Finn then discusses algorithms that present an interface for users to work on, starting from the devoid of purpose Cow Clicker and moving to the interfaces of Uber, Lyft and Amazon's Mechanical Turk, where all algorithmic activity is hidden or abstracted. An important insight concerns the gamification of human relationships and interaction, from the abstraction of ridesharing drivers to everyday activities on social networks.

The fourth and final case examined in the book is automatic algorithmic trading and cryptocurrencies like Bitcoin, where the algorithms perform arbitrage in speed and scale far beyond human capabilities. Thus, they are able to manipulate money or news and eventually generate and become value themselves.

The final chapter describes what the author sees as a major transformation of Habermas' bourgeois public sphere to something akin to an "algorithmic sphere" that is opaque to the wide public and probably not even controllable by the original programmers. Whether this can help us achieve our desires of universal knowledge and perfect self-knowledge is an open question, so the author concludes with a call for creation of and participation in humanistic research that engages directly with algorithmic culture, a novel field that he names "experimental humanities."

The main text of the book is supplemented with notes, an extensive bibliography, and a detailed index.

Finn approaches an admittedly huge topic with courage, usually erring on the side of extended breadth rather than informative depth. His main message is occasionally muddled, but the work as a whole contains some impressive insights and can definitely serve to trigger more thoughts on every reader's mind. What Algorithms Want is a well-written book that covers the fascinating subject of algorithms from a novel perspective. As we move daily towards a society where algorithms determine numerous

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outcomes and change human culture and values, an analytical presentation of this system is much appreciated.

Alexios Zavras, Munich, Germany

Annika Frye, *Design und Improvisation. Prozesse, Produkte und Methoden.* Bielefeld: transcript Verlag, 2017, 258 pages, €29,99.

The topic of this excellent book is the role of improvisation in product design, rather than in engineering design. Annika Frye investigates the development of product design with special reference to Germany after World War II. Inspired by authors such as Charles Jencks and Nathan Silver and their concept of "Adhocism" (1972), in which designers using improvisation rely on resources which are already at hand, Frye makes a strong case for the important role of improvisation in product design as a formative but also problem-solving principle and supports her thesis with many well-chosen und convincing examples.

Improvisation is generally associated with processes that are spontaneous, unpredictable and, in many cases, preliminary. There is a difference between improvisation in product design, when models are built but then revised and developed into a new (serial) product, and improvisation which can still be seen on the object. In the latter cases, which are not intended for serial production, improvisation and product are one and the same process.

In her book, Frye gives a good survey of the development of design theory, which in the 1950s and 1960s often focused on individual designers, frequently admired as "stars", aiming at serial production in large numbers. During that time, design was widely regarded as a planned, closed process. In such a context, there seemed to be no need for improvisation which was usually looked upon as an emergency and makeshift solution and an indication of failure in a well-planned design process. But when concepts such as tinkering and bricolage gained prominence, improvisation started to play an increasingly large role in the theory and practice of product design. However, as Frye, relying on sound evidence, rightly points out, there was much improvisation in product design already in the 1950s and 1960s. This is true, for example, of Robert Ullmann and the design process of Braun's (a

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